

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Instructions

This quiz will use all of the Coursera Lab Jupyter notebooks from this week. I recommend that you have another browser tab or window available and that you open the labs there. You can run the labs in parallel with the quiz, and enter quiz answers from the labs directly.

1.

To compute $\hat{x}_{k|k}$, $\Sigma_{k|k}^+$, and $\hat{z}_k$, what information do we require? Select all that apply.

☒ We need to know model matrices A_{k-1} , B_{k-1} , C_k , and D_k .

Correct

Yes. These matrices are needed in the prediction steps.

☒ We need to know the process-noise covariance matrix Σ_w .

Correct

Yes. We need to know this information for step 1b.

☐ We need to know the sensor-noise covariance matrix Σ_v .

☒ We need to have an estimate of the prior values of the state vector, $\hat{x}_{k-1}^+$.

Correct

Yes. This is needed in step 1a.

☒ We need to know the error covariance matrix of the prior state estimate, $\Sigma_{x,k-1}^+$.

Correct

Yes. This is needed in step 1b.

2.

What is the significance of the state estimation error covariance matrix $\Sigma_{x,k}^+$?

☒ It allows computing error bounds on the state estimate produced by the Kalman filter.

☐ It is needed to compute the output prediction error $\hat{z}_k \equiv z_k - \hat{z}_k$.

☐ It is needed to be able to compute the Kalman gain matrix L_k .

☐ It indicates the uncertainty of the predicted state.

Correct

Yes. This is the primary significance and use of this matrix.

Instructions for the next questions.

3.

[Uses Coursera Lab Jupyter notebook from Lesson 1.4.2]

What is the true velocity of the mass at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

Hint: Time 25 seconds corresponds to $k = 251$ in the Octave code.

0.055395

Correct

Yes. That is correct.

4.

[Uses Coursera Lab Jupyter notebook from Lesson 1.4.2]

What is the measured position $z(k)$ of the mass at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

0.073309

Correct

Yes. That is correct.

Instructions for the next questions

5.

[Uses Coursera Lab Jupyter notebook from Lesson 1.4.3]

What is the position-estimation error at time $t = 25$ seconds? Recall that error is a signed value and is calculated as "true position minus estimated position". Enter your answer with four digits to the right of the decimal point.

0.0048918

Correct

Yes. That is correct.

6.

[Uses Coursera Lab Jupyter notebook from Lesson 1.4.3]

What is the 3σ bound on the velocity error at time $t = 25$ seconds? Enter your answer with four digits to the right of the decimal point.

0.0074352

Correct

Yes. That is correct.

Instructions for the next questions

7.

[Uses Coursera Lab Jupyter notebook from Lesson 1.4.4]

Run the open-loop state-estimator code from the Coursera Lab. What is the 3σ error bound on the position state at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

0.056301

Correct

Yes. That is correct.

8.

[Uses Coursera Lab Jupyter notebook from Lesson 1.4.4]

Run the Kalman-filter state-estimator code from the Coursera Lab (for the case where the initial state is poorly chosen). During the simulation, how many times did the magnitude of the velocity error exceed the 3σ bounds on velocity error?

1

Correct

Yes. That is correct.

Instructions for the next questions

9.

[Uses Coursera Lab Jupyter notebook from Lesson 1.4.5]

Run the code cells in the notebook from the section titled: "Implementing a Kalman filter using incorrect noise variances". What is the fraction of time that the position-estimate error is outside its 3σ error bounds? Enter your answer with two digits to the right of the decimal point.

Hint: You can compute this answer using the following line of code:

```
length(find(abs(x_err(1,:))>bound@Vatone(1,:)))/nt
```

0.36863

Correct

Yes. That is correct.

10.

[Uses Coursera Lab Jupyter notebook from Lesson 1.4.5]

Run the code cells in the notebook from the section titled: "Implementing a Kalman filter with correlated noises". What is the fraction of time that the velocity-estimate error is outside its 3σ error bounds? Enter your answer with three digits to the right of the decimal point.

0.24975

Correct

Yes. That is correct.